EMC404S Viva Memory Pack

Behavioural Emergencies and Toxicology — the 2-day version

How to use this pack (read this first)

This pack is the **compressed, memorisable core** of your full study guide. Every fact here comes from the full guide, where you will find the detail and the references. For a viva, you need three things: (1) a **framework** to start any answer confidently, (2) the **must-say facts** (numbers, sections, doses), and (3) **practice saying it out loud**.

How to remember it (evidence-based techniques):

1. **Say it, don't read it.** Read one box, close the page, and say it out loud as if to the examiner. Speaking is what the viva tests.
2. **Test yourself at spaced intervals.** Revisit each page after about 1 hour, again that evening, and again the next morning. Forgetting and re-retrieving is what makes it stick.
3. **Use the memory hooks.** Each topic has a mnemonic or "story". Say the hook first, then unpack it.
4. **Teach it.** Explain a topic to someone (or to me in a mock viva). If you can teach it, you know it.
5. **Sleep.** Memory is consolidated during sleep. Do not pull an all-nighter before the viva.

The 2-day plan

Time	Day 1 — Behavioural Emergencies	Day 2 — Toxicology + final rehearsal
Morning (3 h)	Pages: Universal answer framework, Law (MHCA, NHA, consent), capacity	Pages: Toxicology framework, CDST poisoning tool, toxidromes, charcoal, emesis
Midday (2 h)	Violent patient, restraint, sedation doses, suicide, abscondment	Specific poisons table, snakes/spiders/scorpions/marine, plants, scombroid
Afternoon (2 h)	Children, older persons, DSM numbers, sexual assault, domestic violence	Re-test **all** of Day 1 from memory (blurt sheets)
Evening (1–1.5 h)	**Mock viva** (with me) on Day 1 topics; re-read only what you got wrong	**Mock viva** on everything; then light review of numbers page; **sleep early**

1. The universal viva answer framework

Almost every behavioural-emergency viva question can be opened with the same structure. Say it, then go into detail.

"Some Officers Can't Leave Their Desks" — S O C L T D

Letter	Meaning	What you say
S	**Safety**	"Scene safety first: stage until SAPS secures the scene, exits, two arm-lengths, never alone, remove weapons."
O	**Organic first**	"I assume an organic cause until proven otherwise: ABCDE, SpO_2 (O_2 if <92%), BP, **glucose (<3.5 treat)**, temperature, head injury, poisoning. The CDST says: *do not assume aggression is due to a psychiatric condition*."
C	**Capacity and risk**	"I assess capacity — Understand, Retain, Weigh, Communicate — and risk to self and others (suicide, violence, abscondment)."
L	**Least restrictive**	"I escalate stepwise: verbal de-escalation → voluntary oral sedation → brief restraint → IM sedation per CDST → mechanical restraint only on SAPS instruction."
T	**Transport and the law**	"If the patient is a danger to self/others, I am **obligated to transport** (CDST) under **MHCA s 9(1)(c)** / NHA s 7(1)(e), with SAPS under **s 40**, to a State emergency centre."
D	**Document and debrief**	"I document capacity, risk, legal basis, de-escalation, restraint (time, method), drugs and response; I debrief."

Toxicology version — Kaswa's ABCDEFG: **A**irway, **B**reathing, **C**irculation, **D**econtamination (with PPE), **E**nhance elimination, **F**ocus therapy (antidote), **G**et help (Poisons Helpline **0861 555 777**). Add: "Treat the patient, not the poison/lab result."

Phrases that impress examiners

- "Behavioural emergency is a **presentation, not a diagnosis**." (RCEM, 2025)
- "**Organic until proven otherwise.**"
- "A refusal form signed by a patient who lacks capacity has **no legal value**." (Evans, Geduld and Stassen, 2018)
- "Restraint must be **necessary, a last resort, proportionate, lawful, brief, monitored, documented**."
- "The CDST is my national prehospital guidance; ECPs are health care providers but **not** mental health care practitioners under the MHCA."

2. Behavioural emergencies — must-know facts

Definition and causes

Behavioural emergency = abnormal behaviour + **risk of harm** + need for **immediate intervention**.

Causes = "OPS": **O**rganic, **P**sychiatric, **S**ituational. Always exclude organic first.

Organic red flags = "ADVICE":

- **A**cute onset (hours–days)
- **D**isoriented / fluctuating consciousness
- **V**ital signs abnormal (fever, tachycardia, hypoxia, hypotension)
- **I**nitial episode after age ~40
- **C**lues: focal signs, head injury, drugs/alcohol, pupils
- **E**yes see things — visual/tactile hallucinations (psychosis usually = auditory)

ABD (acute behavioural disturbance) — "a strong, hot, fast, fearless patient": agitation, unusual strength, pain insensitivity, hyperthermia, sweating, tachycardia, tachypnoea, non-compliance → risk of **sudden cardiac arrest**; sedate early, minimise struggle.

SA numbers (Tilley et al., 2026, Garden Route EMS 2017–2019): 52% of mental health calls = overdose/self-poisoning; 2.8 suicide deaths per month.

Psychopathology definitions (39.3) — say them crisply

Term	One-line definition
Hallucination	Perception **without** a stimulus
Illusion	**Misinterpretation** of a real stimulus
Delusion	**Fixed false belief**, out of keeping with culture, held despite evidence
Mood vs affect	Mood = **subjective** (patient's words); affect = **objective** (what you see)
Flight of ideas	Rapid jumps with understandable links (mania)
Loosening of association	Unrelated jumps (schizophrenia)
Confabulation	Filling memory gaps unknowingly (Korsakoff)
Disorientation order lost	**Time → place → person**
Catatonia	Stupor, mutism, posturing, waxy flexibility, negativism

Mental State Examination — "ASEPTIC"

Appearance & behaviour · **S**peech · **E**motion (mood + affect) · **P**erception · **T**hought (form + content, incl. **suicidal/homicidal ideation**) · **I**nsight & judgement · **C**ognition (consciousness, orientation, attention, memory).

MMSE (Outcome 40) — utility in one breath

"30 points; ≤23–24 suggests impairment. **Useful**: objective, quick, documentable, helps separate organic from functional, supports capacity assessment. **Limitations**: needs a calm cooperative patient, takes 5–10 min, **biased by education, language and culture** (big issue in SA), does **not diagnose delirium** (use CAM/4AT), does **not** assess suicidality, perception or capacity. So: limited routine prehospital use; useful in the stable patient."

Capacity — "U Really Were Clever"

Understand · **R**etain · **W**eigh · **C**ommunicate. Decision-specific and time-specific. **Do not use the GCS for capacity** (Stander, Hodkinson and Dippenaar, 2021).

Interviewing (39.4)

Safety → introduce yourself → **one voice** → calm, short sentences → open then closed questions → active listening → validate feelings → **never argue with or collude with delusions** → **never lie** → offer choices → set limits → ask directly about suicide → collateral history.

3. The law — the numbers you must not get wrong

Mental Health Care Act 17 of 2002 (not "15 of 2004"; in force Dec 2004)

Memory story: "9 is NOW, 32–33–34 is the FORMAL ROAD, 40 is the POLICE."

Section	Hook	Content
s 9(1)(c)	"**9 = now** (emergency)"	Care without consent if, **due to mental illness**, delay may cause: **D**eath/irreversible harm, **H**arm to self/others, **P**roperty damage — "**Dies, Hurts, Property**"
s 9(2)	"**Report + 24**"	Report in writing to **Review Board**; max **24 hours** unless Chapter V application made
s 32	"**32 = Criteria**"	Involuntary if mental illness + (serious harm **OR** financial interests/reputation) **AND** incapable **AND** unwilling
s 33	"**33 = Application**"	By spouse/next of kin/partner/associate/parent/guardian (parent if <18); **health care provider** if they're unwilling/unavailable; seen within **7 days**; examined by **two** mental health care practitioners
s 34	"**34 → 3 days**"	**72-hour assessment**
s 40	"**40 = Force/Police**"	SAPS **must apprehend** a person likely to seriously harm self/others due to mental illness → **State** health establishment; admit ≤**24 h** pending s 33; **s 40(4)** absconded users — SAPS locate, apprehend, return (Form MHCA 25); **s 40(8)** necessary & proportionate force
s 26–27		**Assisted** care: incapable but **not refusing**
s 25		Voluntary care

Key point: ECPs are **health care providers** (can act under s 9) but **not mental health care practitioners** (psychiatrist, doctor, trained nurse, OT, psychologist, social worker) — so they **cannot** do the s 33 examinations; the prehospital role is **assess, treat, transport for evaluation**.

National Health Act 61 of 2003, s 7 — consent for the incompetent adult

Order of substitute consent (s 7(1)(b)): "**Some People Get Additional Siblings**" → **S**pouse/partner → **P**arent → **G**randparent → **A**dult child → **S**ibling. First choice is a **written mandate** or court/law (s 7(1)(a)). **Emergency** (s 7(1)(e)): delay → death/irreversible damage, and patient hasn't refused.

Children (Children's Act 38 of 2005)

- **12 years** + sufficient maturity → may consent to medical treatment (s 129(2)); surgery needs parental assistance.
- **s 129(6)**: hospital superintendent may consent in an emergency.
- **s 110**: professionals **must** report abuse; **any person may** report; police notify DSD within 24 h.
- HPCSA: a child's refusal can be overridden in their **best interests** (para 15.2); best interests paramount (**Constitution s 28(2)**).
- MHCA s 33: involuntary application for <18 by parent/guardian.

Mandatory reporting — "Everyone, immediately, police"

- **SOA s 54**: **any person** who knows/suspects a sexual offence against a **vulnerable person** (child, mental disability) → report **immediately to SAPS**; failure = up to **5 years**.
- **DVA s 2A**: functionaries (incl. health care personnel) → **Form 2** to social worker/SAPS for DV against **child, older person, person with disability**. **s 2B**: any adult must report these.
- Older Persons Act: report abuse of older persons.

HPCSA Booklet 7 (withholding/withdrawing) — 4 lines for the viva

1. Written for **terminal illness**; prohibits euthanasia and **assisted suicide**.
2. A competent adult may refuse **unless their state of mind makes them incapable** (para 8.1.2) — often the case in suicide.
3. **Emergency + uncertainty → start treatment** until a clearer assessment (para 7.1); inform patient later (7.2).
4. Consult, second opinion, involve family, **document** (para 12). A suicide note is **not** an advance directive.

4. Violent patient, restraint and CDST sedation

CDST "Aggression or agitation" tool — learn this as a story

"Talk → Offer → Hold → Inject → Check → Repeat → Watch → Transport"

1. **Talk** (de-escalate). Try: safe distance, quiet area, listen, calm, kindness, boundaries. Avoid: turning back/being cornered, direct eye contact, unnecessary touch, arguing with delusions, reasoning, prolonged talk.
2. **Offer** voluntary sedation (adult): **midazolam 7.5 mg PO** or **10 mg buccal/IN**, diazepam 10 mg PO, lorazepam 2 mg PO; IV: **midazolam 2 mg** (repeat 15 min).
3. **Hold**: **5 people (4 limbs + head) + 1 to inject**; brief, only to facilitate sedation; **mechanical restraint only on SAPS instruction**; never prone with weight on the back/neck.
4. **Inject** (rapid sedation): **midazolam 0.2 mg/kg IM, max 15 mg — >60 years: 0.1 mg/kg**. If no midazolam: **lorazepam 4 mg IM** or ketamine 4 mg/kg IM (ANT/ECP).
   - Hook: "**Two-Fifteen, halve over Sixty; Four-Four backup**" (0.2 mg/kg, max 15; >60 halve; lorazepam 4, ketamine 4).
5. **Check**: primary survey, SpO_2 (O_2 <92%), BP (<90/60 shock), **glucose <3.5** (D50 20 mL / D10 3 mL/kg / glucagon 1 mg IM), head injury, poisoning; IV access.
6. **Repeat** after **15 min (IV) / 30 min (IM/PO)**: **midazolam 0.1 mg/kg IM, max 5 mg** (>60: 0.05 mg/kg) or lorazepam 2 mg IM.
7. **Watch**: BP, HR, RR, SpO_2, GCS **every 10 min**; **check distal pulses** in restrained limbs.
8. **Transport** — **obligated** if: "**SOPPS**" → **S**elf (danger/suicide risk), **O**thers, **P**roperty damage, **P**olice request hospitalisation, **S**edated needing monitoring.

Children <13 (CDST): midazolam 0.2 mg/kg buccal/IN, 0.15 mg/kg IM, 0.05 mg/kg IV, 0.5 mg/kg PO.

RCEM 10 domains of de-escalation (Richmond et al., 2012)

"**Respect, Don't provoke, Establish contact, Concise, Wants, Listen, Agree, Limits, Choices, Debrief**" — hook: "**Real Doctors Explain Calmly; Wise Listeners Agree, Limit, Choose, Debrief.**"

Evidence on sedative agents (Saucedo Uribe et al., 2025; RCEM, 2025)

- RCEM (UK ED): first line **IM ketamine 4 mg/kg or droperidol 5–10 mg**.
- **Haloperidol alone** → frequent dystonia; "difficult to justify" without promethazine; **haloperidol + promethazine** effective with fewer adverse effects.
- **Midazolam** fastest but most respiratory events, worst in elderly. **Lorazepam + antipsychotic** better than either alone.
- **Never haloperidol/droperidol in Parkinson's or Lewy body dementia.**
- Monitor with **capnography ($ETCO_2$)** — detects hypoventilation before SpO_2 falls (especially on O_2).

Ethics of restraint — "NLP-LBMD"

Necessary · **L**ast resort · **P**roportionate · **L**awful · **B**rief · **M**onitored · **D**ocumented (+ never punitive). Four principles: autonomy vs beneficence, non-maleficence, justice.

SA evidence on EMS practice (critical appraisal ammunition)

- **Evans, Geduld and Stassen (2018)**, Cape Town: **80%** no training in psychiatric patients; **7%** MHCA training; providers wrongly believed transport without consent is "kidnap"; relied on refusal forms; negative attitudes ("waste of time"); patients died after being left on scene; 56% of staff assaulted (other study).
- **Stander, Hodkinson and Dippenaar (2021)**: deception used to get patients to go; GCS misused for capacity; SAPS reluctant; MHCA silent on EMS role.

5. Suicide, abscondment, children, older persons

Suicide — numbers "23.5 · 5× · 41 · 80"

- SA suicide rate **23.5/100,000** (global 8.9); 3rd leading cause of death age 15–29 (Tilley et al., 2026).
- Males **5×** more likely to die; men → **hanging**; women → **poisoning/overdose**; peak risk age **41**.
- **80%** of people who die by suicide told someone first; **10%** of those who threaten/attempt eventually die (Evans, Geduld and Stassen, 2018).
- Strongest predictor: **previous attempt**. Others: hopelessness, mental illness, substance use, male, isolation, access to means, recent loss.

Interview = the "FUNNEL" (general → specific): Feelings → thoughts of Death → Suicidal thoughts → Intent → Plan & Means → Preparation/notes → Past attempts → Protective factors. **Ask directly** — asking does not increase risk.

Model questions: "Have you had thoughts of killing yourself?" · "Did you want to die?" · "Do you still want to die now?" · "How do you feel about being alive?" · "Did you leave a note?" · "Where are the tablets now?" · "Are voices telling you to hurt yourself?" · "What has stopped you?"

Management: treat the method (hanging → airway + C-spine, cut away from knot; overdose → CDST poisoning tool), never leave alone, remove means, **transport every attempt** (obligated if suicide risk), document. Death by suicide = unnatural death → SAPS, Inquests Act, preserve scene.

Abscondment (Outcome 47)

Risk: psychosis, no insight, fear, intoxication, suicidal intent, previous abscondment, open environment, delays. Prevent: rapport, honesty, supervision, attendant near door, doors secured, SAPS escort, voluntary sedation. If they run: **don't chase into danger**, notify **SAPS** (description, risk), MHCA **s 40(4)** for users who abscond from facilities (Form MHCA 25).

Children — key hooks

- CDST child = **<13 years**; legal child = **<18**.
- **Mesthrie (2025)**: child homicide rate **~2× global**; **45%** of households report violence against children; CRC **Article 19**; gaps = **participation** and **gender**.
- Pre-adolescents: brief parental holding often works; adolescents behave like adults (RCEM).
- Evans et al.: providers let families decide for suicidal minors — **wrong**; act in the child's best interests.

Older persons — "Delirium until proven otherwise"

- Causes: hypoxia, hypoglycaemia, **infection (UTI, pneumonia)**, stroke/subdural, pain, **urinary retention, constipation**, medications (Beers list), withdrawal.
- **Non-drug first**: one voice, approach from front, glasses/hearing aids, family, comfort, quiet.

- CDST: **halve midazolam if >60** (0.1 mg/kg; repeat 0.05). Avoid benzos except withdrawal/seizures; **no ketamine routinely**; no haloperidol in Parkinson's/Lewy.
- "Start low, go slow, don't stack, calm not comatose" (Desai et al.; RCEM).

6. DSM-5-TR — the number ladder

Disorder	Must-say criteria
Schizophrenia	≥2 of: delusions, hallucinations, disorganised speech, disorganised/catatonic behaviour, negative symptoms (≥1 of first 3); **6 months** total, **1 month** active
Brief psychotic / schizophreniform	**1 day–1 month** / **1–6 months**
Mania (Bipolar I)	**7 days** (or hospitalised); **3** DIGFAST (**4** if only irritable)
Hypomania (Bipolar II)	**4 days**, no psychosis/hospitalisation; Bipolar II = hypomania + major depression, **never mania**
Major depression	**5 of 9** for **2 weeks**, incl. depressed mood or anhedonia (SIG E CAPS)
Panic attack	Peaks within minutes; ≥4 of 13 symptoms — **diagnosis of exclusion** (ECG, glucose, SpO_2)
GAD	6 months, ≥3 symptoms
Acute stress disorder	**3 days–1 month**, ≥9 symptoms
PTSD	**>1 month**; intrusion, avoidance, negative cognition/mood, arousal; first responders count as exposed
Catatonia	≥**3 of 12**; treat **lorazepam** / ECT; avoid antipsychotics
Anorexia	Low weight, fear of weight gain, body image; **highest psychiatric mortality**; **refeeding syndrome** (↓phosphate)
Bulimia	Binge + compensation ≥1/week × 3 months; **hypokalaemic metabolic alkalosis**, Russell's sign
Substance use disorder	**≥2 of 11** in 12 months: mild 2–3, moderate 4–5, severe ≥6
Delirium	**Acute, fluctuating, inattention**, altered awareness; medical cause (CAM)
Intellectual disability	Intellectual + **adaptive** deficits, developmental onset
Autism	Social communication deficits + restricted/repetitive behaviour/sensory issues

Ladder hook: "7-4-2-6" → Mania **7** days, hypomania **4** days, depression **2** weeks, schizophrenia **6** months.

Personality clusters = "Weird, Wild, Worried": A (paranoid, schizoid, schizotypal) · B (antisocial, borderline, histrionic, narcissistic — self-harm & violence) · C (avoidant, dependent, obsessive-compulsive).

Serotonin syndrome vs NMS: serotonin = **hours, clonus/hyperreflexia, diarrhoea** (cyproheptadine); NMS = **days, lead-pipe rigidity, bradyreflexia** (bromocriptine/dantrolene). Both: stop drug, benzos, cool.

SA drugs: tik (meth), Mandrax (methaqualone — sedative **with** hypertonia/hyperreflexia), nyaope/whoonga (heroin mix), dagga, cat.

7. Sexual assault and domestic violence

Sexual Offences Act 32 of 2007

- **Rape (s 3)** = "**UIP-NC**": **U**nlawful, **I**ntentional, sexual **P**enetration (**to any extent**; genital/anal/oral; object, finger) without **C**onsent. **Gender-neutral**; marital rape included.
- **Sexual assault (s 5)** = sexual **violation** without penetration (or inspiring belief you will be violated).
- **Consent** = "voluntary or uncoerced agreement"; **no consent** if: **Force/threats, Abuse of power, Fraud, or Incapable** (asleep, unconscious, **drugged/drunk**, **<12 years**, mental disability) — "**FAFI**".
- Age of consent **16**; <12 cannot consent.
- **s 28**: **PEP at State expense** if charge laid or reported at designated facility **within 72 h**.
- **s 54**: report sexual offences against vulnerable persons **immediately to SAPS** (5 years).

Prehospital care of the survivor — "Life, Listen, Leave it, Label, Leave fast"

1. **Life** before evidence (airway, bleeding, **strangulation** = red flag, head injury).
2. **Listen**: believe, trauma-informed, permission before touch, same-gender provider, minimal questions (time of assault!), survivor's own words in quotes.
3. **Leave it** (evidence): no washing, eating, drinking, smoking, changing; collect urine in a clean container if they must void (drug-facilitated); **don't cut through holes**; **paper bags, one item per bag**; no internal exam.
4. **Label**: seal, label, sign; **chain of custody** to SAPS (name, rank, time).
5. **Leave fast**: transport to **Thuthuzela Care Centre** / designated facility — **PEP within 72 h**.

Injuries: genital (posterior fourchette, "TEARS" = Tears, Ecchymosis, Abrasions, Redness, Swelling); extragenital (grip marks upper arms, inner thighs, neck, bites, defence injuries). **No genital injury ≠ no rape.**

PEP (Horak et al., 2023): **TLD × 28 days** (≥10 y and ≥30 kg); child: AZT+3TC+DTG; HIV tests **baseline, 6 weeks, 3 months**; + emergency contraception, STI prophylaxis, hepatitis B, tetanus.

TCC = NPA (SOCA) one-stop centres — medical, forensic (SAECK, **J88**), counselling, police, prosecution — reduces secondary victimisation.

Domestic Violence Act 116 of 1998 (amended Act 14 of 2021, in force 14 April 2023)

Definition — think "**P-S-E-E-I-H**" + new ones: **P**hysical, **S**exual, **E**motional/verbal/psychological, **E**conomic abuse, **I**ntimidation, **H**arassment (incl. tech tracking) + sexual harassment, related-person abuse, **spiritual** abuse, damage to property, **elder** abuse, **coercive** and **controlling** behaviour, **exposing a child** to DV, unauthorised entry, any other abusive conduct.

Protections — "SAPS-POW!":

- **SAPS** duties (s 2): help, shelter list, medical care, notice of rights.
- **A**rrest at the scene without warrant for violent DV (s 3).
- **P**rotection order — interim (even without notice) → final (ss 5–7); any time incl. after hours; children can apply alone.
- **O**ffence to breach (s 17).
- **W**arrant of arrest — suspended, issued with every order (s 8); weapons seized (s 9).
- Reporting: s 2A functionary Form 2; s 2B any adult (child, disabled, older person).

SA prevalence: ~**21%** of ever-partnered women physical IPV lifetime (SADHS 2016); **>50%** of murdered women killed by intimate partner (Abrahams et al., 2013).

Screening: WHO (2013) — **no universal screening**; ask **selectively and privately** when indicators exist (HITS/HARK questions). USPSTF (2018) supports screening women of reproductive age. **Prehospital answer: selective private inquiry + screen the home for children/older/disabled persons.**

Referral numbers: SAPS **10111**, GBV Command Centre **0800 428 428**, Childline **116**, Poisons **0861 555 777**.

8. Toxicology — must-know facts

Routes (58) — "I-I-A-I": Ingestion (most common), Injection (fastest IV), Absorption (skin/eyes — organophosphates through intact skin!), Inhalation (CO — SpO_2 falsely normal).

Epidemiology (59) — SA numbers

- **Tygerberg PIC** (Veale et al., 2013): non-drug chemicals **52.7%** (pesticides most common), medicines **35.2%** (**paracetamol** most common), biological **12.6%**; adults **61%** intentional; children **99%** accidental.
- **Red Cross Children's** (Balme et al., 2012): **paraffin** commonest (**24%**).
- Pesticides: organophosphates, **aldicarb "Two-step"** (street rat poison). Notifiable: lead, mercury, pesticide poisoning.

CDST Poisoning tool — "Stop, Give, Wash, Coal?, Drip, Sugar, Ask, Identify, Call, Decide, Watch"

- **O_2** if SpO_2 <92%, RR >30 or <9, GCS ≤14.
- **Wash** skin with soap and water (PPE); **eyes 20 min**.
- **Activated charcoal 1 g/kg** (adult 50–100 g) only if "**FOHOA**": **F**ully conscious, **O**ral, **H**armful, within **O**ne hour, **A**dsorbable. **NOT** for: household products, **paraffin/petrol**, **corrosives**, **iron**, **lithium**, **alcohol**. **No milk, no vomiting.**
- Fluids: adult 250–500 mL if SBP <90; child 10 mL/kg.
- Glucose <3.5: D50 20 mL / D10 3 mL/kg / glucagon 1 mg (child 0.5).
- **Poisons Helpline 0861 555 777.**
- Transport if toxic dose, symptomatic, or unknown substance; monitor every 10 min.

Emesis (61) — "Never" (practically)

Ipecac: no outcome benefit, delays charcoal/antidotes, causes aspiration (Höjer et al., 2013); CDST: **avoid inducing vomiting**. Contraindications: ↓consciousness, seizure-risk drugs, corrosives, hydrocarbons, sharps, infants <6 months. Only theoretical if **all 5**: no contraindication, serious toxicity, no alternative, >1 h from care, within 30–90 min.

Charcoal appraisal (62.2) — one breath

"Adsorbs most drugs, best within **1 hour**, cheap, safe if alert; MDAC for carbamazepine, phenobarbital, dapsone, quinine, **theophylline**. **But** no trial shows reduced mortality, aspiration risk, useless for many SA poisons (paraffin, corrosives, iron, lithium, alcohols). So: **selective**, never compromise airway or delay transport."

Toxidromes (62.3) — the pupil + skin method

	Pupils	Skin	Key sign
Sympathomimetic	Big	**Wet** (sweaty)	Hot, fast, agitated
Anticholinergic	Big	**Dry**, red	Retention, no bowel

	Pupils	Skin	Key sign
			sounds, delirium — "hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter, full as a flask"
Opioid	**Pinpoint**	Normal	**Respiratory depression**
Cholinergic	**Pinpoint**	**Wet**	**Secretions**, bradycardia — **DUMBELS** (Diarrhoea, Urination, Miosis, Bradycardia/Bronchorrhoea, Emesis, Lacrimation, Salivation)
Sedative-hypnotic	Normal/small	Normal	Slurred speech, ↓LOC
Serotonin	Big	Wet	**Clonus**, hyperreflexia, diarrhoea

Rule: "Big + wet = stimulant; big + dry = anticholinergic; small + dry = opioid; small + wet = cholinergic."

CDST antidotes — the dose list

Situation	Adult	Child
Opioid (resp. depression)	*Naloxone **0.4 mg*** IV/IM (**0.04 mg** in known abuser) → **0.8** → **2** → **4 mg** q2min, max **10 mg** total	**0.1 mg/kg** (max 2 mg/dose) q2min, max 10 mg
Organophosphate	^Atropine **2 mg** IV^, **double every 3 min** until chest clear, then **20% of total per hour**; no succinylcholine	**0.05 mg/kg**
TCA (wide QRS, VT, seizures, hypotension)	^Sodium bicarbonate 8.4% **100 mL** over 3 min^, repeat once; if intubated hyperventilate, $ETCO_2$ **20–25**	**2 mL/kg**
BB/CCB hypotension	Fluids → ^Ca chloride 10% **10 mL** or Ca gluconate 10% **30 mL** over 10 min^ → ^adrenaline infusion^ (1	Ca chloride 0.2 mL/kg / gluconate 0.6 mL/kg; *glucagon **100 mcg/kg*** if BB

Situation	Adult	Child
	mg/50 mL, start 5 mL/h)	
Stimulant chest pain	Aspirin **300 mg**, ^diazepam 5 mg / midazolam 1 mg IV^, ^ISDN 5 mg / GTN 0.4 mg SL^	
Hyperthermia ≥38 °C	Undress, water spray, fan/air-con	Same
Alcohol / malnourished	Thiamine **100 mg** IV/IM	

9. Specific poisons — one line each

Poison	Mechanism hook	Danger sign	Key management
Paracetamol	**NAPQI**, glutathione depleted	**Looks well for 24 h**	>150 mg/kg toxic; charcoal <1 h; level at **≥4 h**; **NAC 150 / 50 / 100** mg/kg (1 h / 4 h / 16 h); **transport all**
TCA	Na^+ channel block + anticholinergic	QRS **>100** → **seizures, >160** → **VT**; crashes fast	Bicarbonate; no flumazenil/physostigmine
Salicylate	Uncouples oxidative phosphorylation	**Tinnitus**, hyperventilation, hyperthermia; **resp. alkalosis + high-AG acidosis**	Never hypoventilate after intubation; urine alkalinisation; dialysis
Iron	Corrosive + mitochondrial toxin	**Latent phase** looks better; >60 mg/kg serious	**No charcoal**; fluids; WBI; deferoxamine
Lithium	Narrow TI, renal excretion	Coarse tremor, confusion, ataxia; chronic worse	**No charcoal**; saline; **dialysis**
Sulphonylureas	Insulin release	**Delayed, recurrent hypoglycaemia**; 1 tablet kills a toddler	Dextrose; transport all; octreotide (hospital)
Organophosphate / aldicarb	AChE inhibition	DUMBELS + fasciculations; death from **respiratory failure**	PPE, decontaminate, **atropine to dry chest**, airway
Carbon monoxide	Binds Hb **200–250×**, left shift	Headache, whole family, winter braziers; **SpO_2 normal**	**100% O_2** (half-life 4–6 h → ~1 h); HBO if severe
Cyanide	Blocks cytochrome oxidase	House fires, mining; **lactic acidosis**, seizures, collapse	100% O_2; **hydroxocobalamin 5 g** (70 mg/kg child) in hospital
Chlorine	Forms HCl/HOCl in airway	Cough, stridor, **delayed pulmonary oedema**	Upwind, fresh air, O_2, salbutamol, airway
Caustics (alkali)	**Liquefaction**, oesophagus	Drooling, stridor	No vomiting/charcoal/neutralising; airway

Poison	Mechanism hook	Danger sign	Key management
Acids	**Coagulation**, stomach	Haematemesis, acidosis	Same
HF acid	Fluoride binds **Ca/Mg**	**Pain out of proportion**, VF	Irrigate, **calcium gluconate gel**, IV calcium
Paraffin / hydrocarbons	Surfactant destruction	Cough/choke → **aspiration pneumonitis**	**No vomiting, no charcoal**; O_2; observe ≥6 h; inhalants → don't startle, avoid adrenaline
Ethylene glycol / methanol	ADH → toxic acids	Drunk without smell; high AG acidosis; **AKI** (EG) / **blindness** (methanol)	No charcoal; fomepizole/ethanol; dialysis
Colchicine / flame lily	Microtubules	GI → multi-organ failure → alopecia	Transport all; no antidote
Theophylline	Adenosine antagonism	**Intractable vomiting**, tachyarrhythmia, refractory seizures, ↓K	MDAC, dialysis
Ethanol	GABA↑, NMDA↓	**Hypoglycaemia** (children), hypothermia, hidden head injury	Airway, glucose, thiamine
NSAIDs	COX inhibition	Mostly benign; mefenamic acid → seizures	Supportive
Psychiatric meds	Antipsychotics: sedation, hypotension, QT	Serotonin syndrome vs NMS	Benzos, cooling
DFSA	Alcohol (most common), GHB, benzos	Amnesia, sudden "drunkenness"	Collect **first urine**; treat as sexual assault
Chemical warfare	Nerve agents = organophosphates		**Upwind, undress, decontaminate**, atropine + oxime + benzo; phosgene → **rest**

10. Plants, traditional medicine, bites and stings, scombroid

Plants and mushrooms — "Datura dries, Oleander slows, Impila kills the liver, Death cap waits"

- ***Datura stramonium*** (thorn apple, "malpitte") → **anticholinergic** (benzos, cooling; physostigmine in hospital).
- **Oleander, slangkop (*Drimia*), climbing onion (*Bowiea*), plakkie → cardiac glycosides** (bradycardia, AV block, ↑K; digoxin Fab).
- ***Callilepis laureola*** **(Impila) → liver and kidney failure**, hypoglycaemia, especially children (enemas).
- ***Amanita phalloides*** **(death cap) → GI symptoms >6 h after eating**, false recovery, **liver failure** day 2–5.
- *Senecio* → veno-occlusive liver disease; *Ricinus* (castor) → ricin; *Dieffenbachia* → oral swelling; flame lily → colchicine.

Traditional medicine (63.2) — "Pesticide, Liver, Heart": pesticides in muthi → **cholinergic crisis**; hepatotoxicity → **Impila**, pyrrolizidines; cardiovascular collapse → **glycoside plants, enemas, potassium dichromate**. Ask respectfully; bring the remedy; check glucose; transport.

Snakes — "Swell, Weak, Bleed"

Syndrome	Snakes	Signs	PIB?
Swell (cytotoxic)	**Puff adder**, Gaboon adder, **Mozambique spitting cobra**	Pain, progressive swelling, necrosis, shock	**NO**
Weak (neurotoxic)	**Mambas**, **Cape/snouted/forest cobra**, (rinkhals, berg adder)	**Ptosis** → bulbar → **respiratory paralysis**	**Yes** (if long transport)
Bleed (haemotoxic)	**Boomslang**, vine snake	**Delayed** bleeding	No

First aid: reassure, keep still, **remove rings**, splint, mark swelling edge, IV in other arm, analgesia (no NSAIDs), **ventilate neurotoxic paralysis** (BVM saves lives), **irrigate spat venom in eyes**. **Never**: cut, suck, tourniquet, ice, shock.

Global: **1.8–2.7 million** envenomings, **81–138 thousand deaths**/year; WHO neglected tropical disease (2017).

Antivenom (SAVP polyvalent) covers 10 — "**3 mambas, 3 cobras + the spitter, 2 adders, 1 rinkhals**": black, green, Jameson's mamba; Cape, forest, snouted cobra; Mozambique spitting cobra; puff adder, Gaboon adder; rinkhals. **Not**: berg adder, **boomslang** (separate monovalent), vine snake (none). **IV only, same dose for children**, adrenaline ready. Reactions: **early anaphylactic** (adrenaline 0.5 mg IM), **pyrogenic**, **serum sickness at 5–24 days**.

Spiders and scorpions

- **Button spider** (*Latrodectus*) → **neurotoxic**: severe pain, **board-like abdomen**, cramps, sweating → analgesia, benzos, antivenom.
- **Violin spider** (*Loxosceles*) → **necrotic ulcer**; sac spider → small necrosis.
- **Scorpion rule**: **"Thick tail, thin pincers = danger"** (*Parabuthus*) → neurotoxic: pain, paraesthesia, drooling, dysphagia, ptosis, **respiratory failure** in children/elderly → ventilate; antivenom.

Marine

- **Bluebottle**: rinse with **seawater**, **hot water 45 °C × 20 min** (no vinegar for *Physalia*).
- **Stonefish/stingray**: **hot water** 30–90 min (heat-labile venom), analgesia, X-ray.
- **Red tide → paralytic shellfish poisoning** (west coast mussels), **pufferfish** (tetrodotoxin), **cone snail → paralysis → ventilate**.

Scombroid (65) — "Hot fish, hot face"

Improperly chilled tuna/mackerel/sardines → bacteria turn **histidine → histamine**; **heat-stable** (cooking doesn't help). **10–30 min**: **flushing**, urticaria, **peppery/metallic taste**, headache, palpitations, GI upset, hypotension; **several diners**; **not an allergy**; isoniazid worsens it. **Treat**: ABCDE, **adrenaline 0.5 mg IM** if anaphylaxis-like, O_2, fluids, salbutamol; **H1 + H2 antihistamines** at facility; resolves in 12–24 h; **notify environmental health**, keep the fish.

11. The 40 most likely viva questions (self-test list)

Cover the right column, answer aloud in 1–2 minutes, then check the must-say keywords.

#	Question	Must-say keywords
1	What is a behavioural emergency?	abnormal behaviour, risk of harm, immediate intervention, presentation not diagnosis
2	Compare organic, psychiatric and situational causes	OPS, ADVICE red flags, glucose, organic first
3	Define hallucination, illusion, delusion	no stimulus / misinterpretation / fixed false belief
4	Mood vs affect	subjective vs objective
5	Headings of the MSE	ASEPTIC
6	Utility of the MMSE prehospitally	30 points, objective, but cooperation, culture/language bias, no delirium (CAM)
7	How do you assess capacity?	URWC, decision-specific, not GCS
8	How do you interview an agitated patient?	safety, one voice, don't argue with delusions, don't lie, ask about suicide
9	Principles of prehospital care in behavioural emergencies	SOCLTD
10	CDST sedation for an aggressive adult	midazolam 0.2 mg/kg IM max 15, >60 0.1, lorazepam 4, ketamine 4, repeat 0.1 max 5
11	How do you restrain safely?	5+1, brief, supine, no neck/chest pressure, SAPS for mechanical, pulses, 10-min obs
12	Ethics of chemical restraint in psychosis	autonomy vs beneficence, NLP-LBMD, evidence (haloperidol dystonia, midazolam resp.)
13	When are you obliged to transport?	SOPPS (CDST)
14	MHCA s 9	due to mental illness, Dies/Hurts/Property, Review Board, 24 h

#	Question	Must-say keywords
15	MHCA s 40	SAPS must apprehend, State facility, 24 h, absconders, proportionate force
16	Involuntary admission pathway	s 32 criteria, s 33 application (7 days, 2 practitioners), s 34 72 h
17	Why can't an ECP admit involuntarily?	not a mental health care practitioner
18	Who consents for an incompetent adult?	written mandate, Spouse-Parent-Grandparent-Adult child-Sibling, emergency s 7(1)(e)
19	How can EMS violate rights of the mentally ill?	unlawful detention, assault, discrimination, deception, confidentiality, abandonment
20	De-escalation techniques	RCEM 10 domains, CDST try/avoid
21	Safety measures when violence anticipated	stage, SAPS, exits, never alone, remove weapons, doors, code word
22	Suicide risk factors & SA statistics	23.5, 5×, 41, 80%, previous attempt, hopelessness
23	Questions to determine suicidal intent	funnel, direct questions, intent/plan/means/note/regret
24	Patient attempted suicide and refuses — what now?	capacity, s 9(1)(c), CDST obligation, SAPS, refusal form invalid, Evans study
25	HPCSA Booklet 7 and suicide	terminal illness, para 8.1.2 exception, para 7.1, no assisted suicide
26	Abscondment risk and management	risk factors, prevention, don't chase, SAPS, s 40(4)
27	Behavioural emergencies in children — law	<13 CDST, 12-year consent, s 129(6), s 110, SOA s 54, best interests
28	Agitation in older persons	delirium, causes, non-drug first, halve midazolam, avoid haloperidol in Parkinson's
29	Bipolar I vs II	7 days/4 days, mania vs

#	Question	Must-say keywords
		hypomania + MDE
30	Serotonin syndrome vs NMS	hours/clonus vs days/rigidity
31	Definition of rape and sexual assault	UIP-NC, violation, consent FAFI, gender-neutral
32	Evidence preservation	no washing etc., paper bags, no cutting holes, chain of custody
33	PEP	72 h, TLD 28 days, tests 0/6 weeks/3 months, s 28 State expense
34	Definition and protections of the DVA	definition list, SAPS-POW
35	Should EMS screen for DV?	WHO selective inquiry, privacy, environment screening
36	Routes of poisoning	IIAI, PPE, skin wash, eyes 20 min
37	Toxidromes	pupil + skin rule, DUMBELS, anticholinergic rhyme
38	Activated charcoal — appraise	FOHOA, exclusions, no outcome evidence, selective
39	Role of emesis	no; ipecac; contraindications; 5 conditions
40	Snakebite management & antivenom	Swell/Weak/Bleed, PIB only neuro, ventilate, 10 species, reactions

Bonus: if you blank, use this rescue line: “Let me approach this systematically — safety, organic causes first, capacity and risk, least restrictive management according to the CDST, the legal framework, and documentation.” Then fill in what you know. Examiners reward structure.